

Reg. No.

Third Semester BCA (SEP) Degree Examination
Dec./Jan. 2025-26
(Regular)

INTERNET OF THINGS (IOT)

Time : 1 Hrs

Max. Marks : 40

Instruction to Candidates: 40 objective type questions carrying 1 mark each.

1. Which of the following is not on IOT communication model?
~~(a)~~ Device-to-Derive (b) Device-to-Cloud (c) Cloud-to-Device (d) Device-to-Satellite
2. What is the primary function of an IoT Gateway?
(a) Collect Sensor Data ~~(b)~~ Translate Communication Protocols
(c) Store data for later retrieval (d) Process user commands
3. Which IoT template is related to the application layer where user data applications are processed?
(a) Level - 1 (b) Level - 2 ~~(c)~~ Level - 4 (d) Level - 5
4. Which type of IoT device would typically use a low-power, wide-area network (LPWAN)?
(a) Mobile phone ~~(b)~~ Smart meter (c) Laptop (d) Desktop Computer
5. Which of the following is NOT part of the IoT's enabling technologies?
(a) RFID (b) Bluetooth (c) AI ~~(d)~~ SMS
6. What type of Data is typically transmitted in an IoT?
(a) Sensory data (b) User Input dates
(c) System Status information ~~(d)~~ All of the above
7. Which IoT protocol is designed for web series in constrained environments?
(a) HTTP ~~(b)~~ CoAP (c) FTP (d) MQTT
8. Which layer in the IoT architecture is responsible for the user interface and interaction?
(a) Perception Layer (b) Application Layer
(c) Network Layer ~~(d)~~ Business Layer
9. What is the main advantage of using REST API's in IoT communication?
(a) They are lightweight and easy to integrate
(b) They provide scheme communication
(c) They support real-time data transmission
~~(d)~~ They are use for data storage
10. Which of these is an IoT protocol used for continuos streaming of data?
(a) HTTP (b) MQTT ~~(c)~~ Web Socket (d) CoAP

11. Which IoT communication model allows multiple devices to communicate directly with each other without needing a cloud?
 - (a) Device-to-Device
 - (b) Cloud-to-Device
 - (c) Device-to-Gateway
 - (d) Cloud-to-Cloud
12. Which IoT level includes the "Perception" and "Network" layers?
 - (a) Level - 1
 - (b) Level - 2
 - (c) Level - 3
 - (d) Level - 4
13. Which of the following protocols is typically used for resource - constrained device in IoT?
 - (a) HTTP
 - (b) MQTT
 - (c) CoAP
 - (d) FTP
14. Which of the following is a fundamental requirement for IoT communication?
 - (a) Security
 - (b) High Bandwidth
 - (c) Low Power Consumption
 - (d) Low Latency
15. Which IoT layer is most responsible for performing computations and analysis on collected data?
 - (a) Perception Layer
 - (b) Network Layer
 - (c) Application Layer
 - (d) Processing Layer
16. Which of the following is an IoT enabling technology for real-time data analysis?
 - (a) Edge Computing
 - (b) Blockchain
 - (c) Cloud Computing
 - (d) Data Analytics
17. Which IoT application domain typically benefits from energy-saving algorithms and resource management system powered by SDN?
 - (a) Retail
 - (b) Agriculture
 - (c) Energy
 - (d) Logistics
18. Which IoT domain primarily benefits from real-time monitoring of devices like heart rate monitors and fitness trackers?
 - (a) Industry
 - (b) Agriculture
 - (c) Health & Lifestyle
 - (d) Energy
19. Which of the following best describes the role of NFV (Network Function Virtualization) in IoT?
 - (a) It reduces the cost of deploying hardware for network
 - (b) It connects IoT devices directly to the cloud
 - (c) It stores IoT data in decentralized networks
 - (d) It ensure that devices are connected to only one network
20. Which domain uses IoT systems for better waste management, traffic control, and resource optimization in urban environment?
 - (a) Cities
 - (b) Energy
 - (c) Retail
 - (d) Agriculture
21. Which technology is responsible for managing and provisioning virtualized network functions in a more flexible and scalable way to IoT
 - (a) NFV
 - (b) CoAP
 - (c) Bluetooth Low Energy
 - (d) Z-Wave

22. In which IoT domain do sensors collect data from industrial machinery to predict maintenance needs?
 (a) Health & Lifestyle (b) Industry (c) Retail (d) Energy
23. Which of the following is a benefit of using IoT in the retail domain.
 (a) Increased energy consumption.
 (b) Real-time inventory tracking and customer behaviour analysis.
 (c) Reduced data security
 (d) Slower supply chain operation.
24. Which of the following describes a key feature of M2M communication in the agriculture domain?
 (a) Provides remote weather forecasting
 (b) Monitors livestock health through wearables
 (c) Manages social media content for farmers
 (d) Tracks GPS location of farm equipment
25. Which of the following is NOT a typical application of IoT in Home Automation?
 (a) Monitoring and controlling Home appliances
 (b) Smart thermostats for temperature regulations
 (c) IoT - based video surveillance systems
 (d) Agricultural drone management.
26. What is the role of IoT in Energy management?
 (a) Energy consumption monitoring and real-time adjustment for efficiency.
 (b) Real-time communication between devices using wi-fi.
 (c) Only used for weather forecasting.
 (d) Remote controlling of power plants
27. Which of the following IoT technologies is crucial for the logistics domain for trucking shipments?
 (a) NFC (b) GPS (c) RFIO (d) Zigbee
28. Which component is typically responsible for powering an IoT device like Raspberry Pi?
 (a) Solar Panel (b) AC Power Source
 (c) Battery or USB power (d) Ethernet cable
29. Which of the following is a key benefit of the forest Fire Detection IoT System?
 (a) It Provides real-time traffic updates.
 (b) It monitors the health of trees.
 (c) It detects forest fires early using temperature and smoke sensors
 (d) It prevents illegal logging activities.
30. Which of the following sensors is most commonly used to detect smoke in a Forest Fire Detection system?
 (a) Temperature sensor (b) Smoke sensor
 (c) Proximity sensor (d) Humidity sensor

31. Which programming language is often used with Raspberry pi to interface with hardware components like sensors and actuators?
 (a) Java Script (b) Java (c) Python (d) Ruby
32. Which of the following is a characteristic of smart lighting in IoT system?
 (a) Lights that adjust based on sound levels.
 (b) Lights that change color based on user preferences
 (c) Lights that automatically turn off when no one is in the room.
 (d) Lights that respond to user voice commands only.
33. In a Home Intrusion Detection IoT system, which of the following could trigger an alert?
 (a) Motion detected by a camera (b) Unusual temperature readings
 (c) Sound detected by a microphone (d) All of the above
34. Which of the following is an important design consideration for an IoT weather reporting Bot?
 (a) Local data storage for backup
 (b) High power consumption for sensors
 (c) The bot should provide real-time update on weather forecasts
 (d) Frequent manual intervention for data collection.
35. In an air pollution monitoring system, how is air quality typically measured?
 (a) By measuring temperature fluctuations (b) By detecting particles in the air
 (c) By measuring wind speed (d) By recording humidity levels
36. What type of communication protocol would typically be used to connect Raspberry pi to other IoT devices in a Smart Home System?
 (a) Bluetooth (b) MQTT (c) Zigbee (d) All of the above
37. Which of the following describe the working principle of a smart parking system using IoT?
 (a) Using GP3 to track cars in parking lots.
 (b) Sensors in parking spaces detect vehicle presence and communicate availability.
 (c) Using a mobile app to find available spaces in real-time.
 (d) All of the above
38. Which sensors is commonly used in a smart lighting IoT system to detect ambient light?
 (a) PIR sensor (b) LDR (Light Dependent Register)
 (c) Proximity sensor (d) Co2 sensor
39. What is the primary purpose of a forest fire detection system in IoT?
 (a) To monitor wildlife in forests (b) To reduce carbons emissions
 (c) To detect and alert about forest fires early (d) To track forest logging activity
40. Which of the following can trigger an alert in a Home Intrusion Detection IoT system?
 (a) Movement detection in a room (b) Window or door opened without authorization
 (c) A loud noise or glass breakage (d) All of the above